



New iPhone 2018

Apple's budget-offering, iPhone SE, for 2018 will be releasing in May or June according to the regulatory reports. <https://digit.in/newiphone>



Control a drone with your body

The Flyjacket exosuit designed in Switzerland allows you to control a drone according to your body movements. <https://digit.in/flyjacket>

Nanoparticle based Holographic Data storage

80 times thinner and 1000 times more data than your average DVD

Mithun Mohandas | mithun@digit.in

S

torage space is something we can rarely have enough of. The more storage space you get, you

tend to become less frugal and use it all up. We clearly remember the days when a few MegaBytes of RAM were enough up until six months later when the new 256 MB sticks came out and all applications started needing more

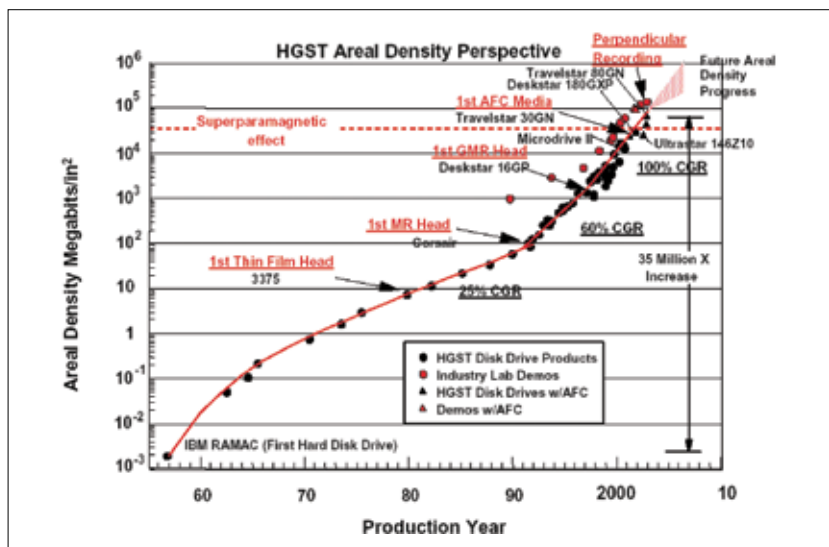
RAM than what you had. Maybe you didn't connect with that example, how about the more recent example of smartphones where 64 GB has become the new "norm", up from 16 GB about 2 years back. Suffice to say, we can never have enough storage space but there's a new technology on the horizon which might just satiate our hunger for more storage space. It's made of nanoparticle-based films that are more than 80 times thinner than a strand of hair and it might just enable us to store

1000 times more data than a DVD in a 10x10 cms piece of film. Also, did we mention that it holographically archives data?

EVOLUTION OF STORAGE TECHNOLOGY

Ever since IBM shipped the first hard drive in the RAMAC system back in 1956 at USD 10,000 / MB of data, the need to have more has always been there. Often, we've only seen incremental changes in storage technology interspersed by huge improvements. However, the overall improvement in areal density hasn't improved by leaps and bounds.

The late '50s and early '60s show a significant spurt in areal density before slowing down till the mid '90s. That's when the first MR (Magnetoresistance) head came to be that allowed for higher density. Then there was GMR (Giant Magnetoresistance) head which helped in a small way and if you've bought your hard drive in the last couple of years then you drive uses something called Perpendicular Magnetic Recording or PMR+. The more recent technology to be implemented in mass production of hard drive is SMR (Shingled Magnetic Recording)



That's a fairly linear growth over the last 60 years